Prohori: Progress Report

OpenCV AI Competition 2026 — AWS Compute Grant check-in

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1 Project

Prohori measures concrete crack width in millimetres from a phone photo. The user places a printed ChArUco board next to the crack; OpenCV uses the board to correct for camera angle and scale and reports the width with an uncertainty range. When a photo cannot support a reliable measurement, the app tells the user how to retake it.

Live app: `https://184-194-241-186.sslip.io` (tap *Demo samples* to try it without a board).

2 Progress

- **Measurement pipeline:** ChArUco detection, `solvePnP` pose, rectification to a known px/mm, crack location, and per-crack width with uncertainty (about 1.5 s per 12 MP photo).
- **Quality gate:** a learned check, calibrated with conformal risk control, that certifies a width only when the photo supports it, and attaches a 90% range to each width.
- **Capture agent:** after each photo, an Amazon Bedrock agent (Nova Lite) chooses the next step (accept, move closer, hold steadier, fix lighting, reposition the board, or stop) from OpenCV's measurements. A free rules policy is the default.
- **Phone web app, history and alerts:** per-joint visit history, change detection beyond measurement uncertainty, engineer alerts with an approval step, and automatic face blurring (OpenCV YuNet) before any photo is stored.
- **Testing:** 206 automated tests passing.

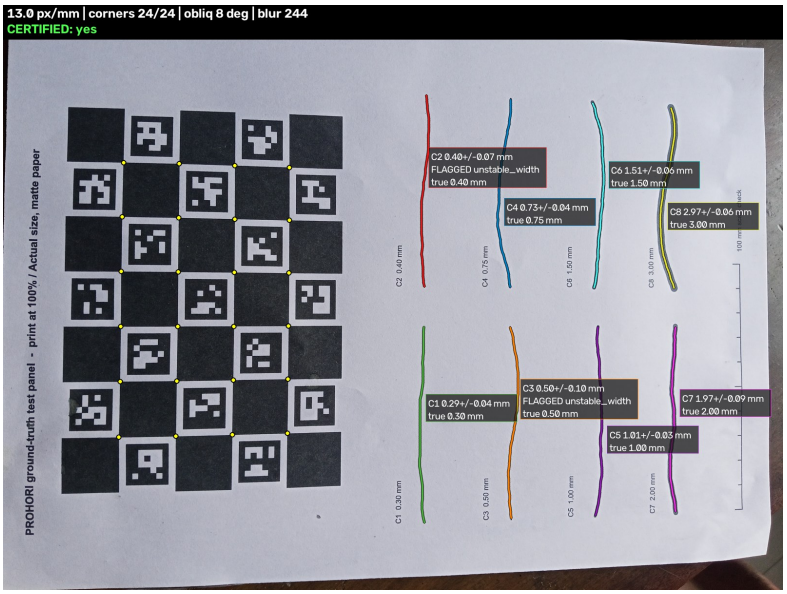


Figure 1: A phone photo processed by Prohori. The board corners set the scale; each crack is labelled with its measured width next to the true printed width (e.g. 0.29 mm vs 0.30, 1.01 mm vs 1.00, 2.97 mm vs 3.00).

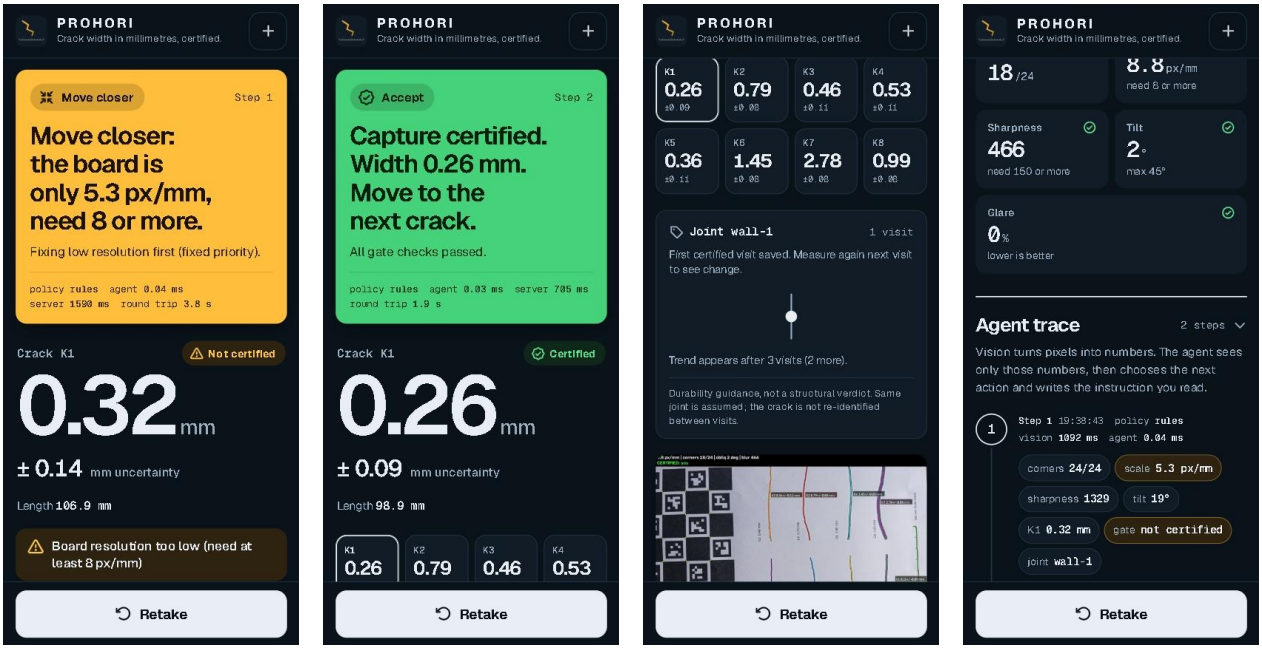


Figure 2: The live phone app. Left to right: photo taken too far away (move closer); certified width after moving closer; all cracks and joint history; photo quality and agent trace.

3 Data

Field ground truth for crack width is hard to obtain, so we built our own test panel with known widths and supplemented it with public datasets.

Dataset	Source	Size	Used for
Ground-truth test panel	Our own. An A4 sheet with the ChArUco board (140×100 mm) and eight printed cracks of exact width (0.30, 0.40, 0.50, 0.75, 1.00, 1.50, 2.00, 3.00 mm), generated by our script, printed at 100% scale on matte paper and photographed with a 12 MP smartphone on 19 Oct 2026 at varied distances and angles.	29 photos, 168 crack measurements	Width accuracy (Fig. 1, results table)
Degraded captures	Generated from the panel photos with controlled blur, motion blur, noise, JPEG compression, glare, extra tilt and lower resolution.	2,584 captures, 18,683 measurements	Quality-gate calibration, 90% ranges (Fig. 3)
Simulated capture problems	Our capture simulator; its distance, tilt, blur and corner-count distributions are fitted to the real panel photos.	100 held-out cases	Agent vs rules test (Fig. 4)
krkCMd	Jakubowski & Tomczak, *Scientific Data* 12:165 (2025); Zenodo, CC BY 4.0. High-resolution scans of real concrete cracks with operator-measured widths.	19,098 width profiles	Resolution vs accuracy limits
Public crack segmentation sets	Crack500, DeepCrack, TUT, CrackSeg9k, CFD, AEL, CrackTree200, GAPs384, Facade390, Road420, from the authors' official releases.	10 datasets	Training the learned crack-segmentation model; real crack shapes for synthetic tests

4 Results so far

Width results are on the ground-truth panel photos; the agent result is on the simulated capture problems (Section 3):

Average width error, certified cracks (held-out photos)	0.048 mm
True width inside the stated 90% range (held-out photos)	92.9%
Capture agent recovery on multi-problem captures (vs fixed rules)	85% (62%)

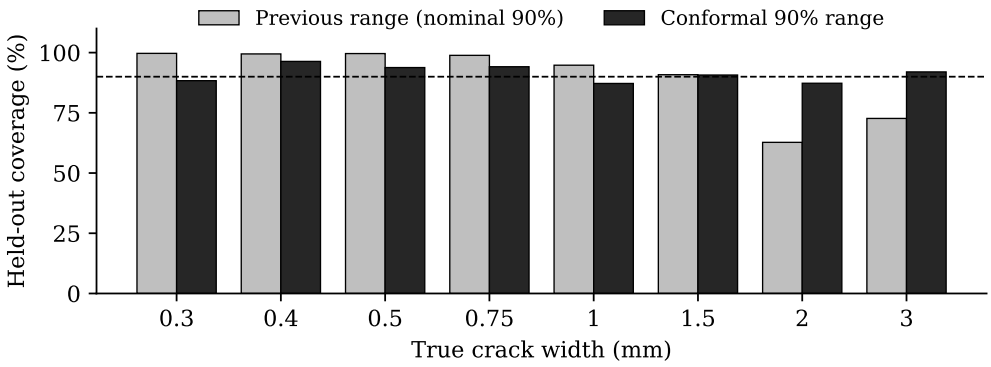


Figure 3: How often the stated 90% range contains the true width, by crack width, on held-out data. The conformal range stays near 90% at every width.

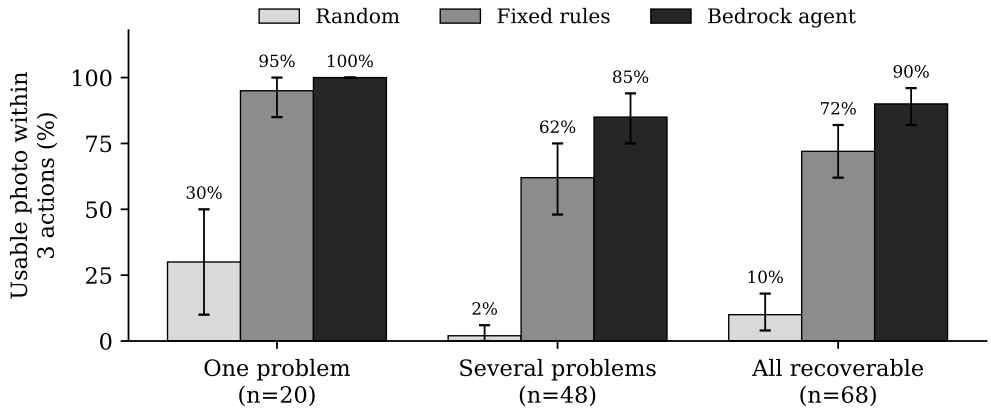


Figure 4: Capture agent vs fixed rules on 100 held-out simulated capture problems (pre-registered test, 95% confidence intervals).

5 AWS usage

Prohori runs as a production service in `us-east-1`:

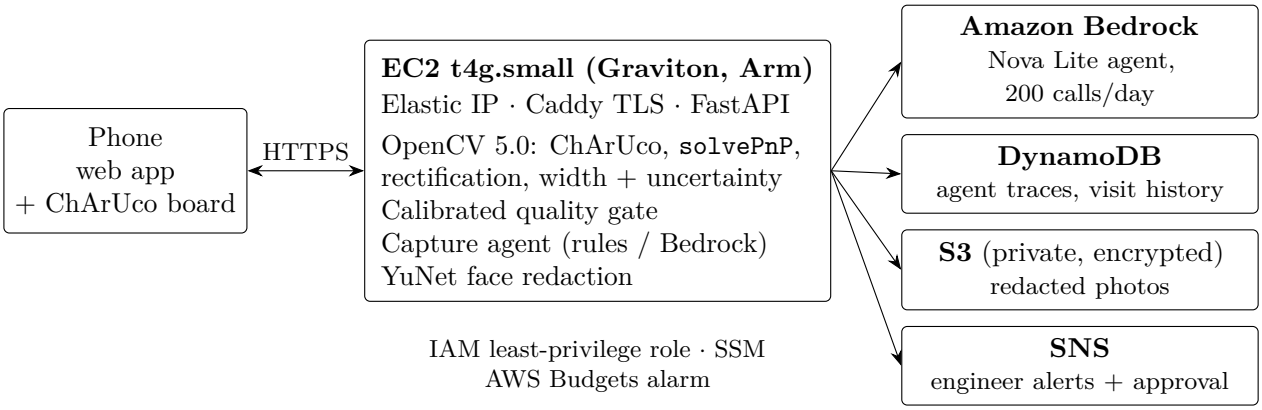


Figure 5: Prohori architecture on AWS (`us-east-1`).

- **EC2 Graviton t4g.small** running OpenCV 5.0 behind HTTPS (Elastic IP, Caddy).

- **Amazon Bedrock** (Nova Lite) for the capture agent, capped at 200 calls per day.
- **DynamoDB** for agent traces and visit history; **S3** (private, encrypted) for photos.
- **SNS** for engineer alerts; **IAM**, **SSM** and an **AWS Budgets** alarm.

The service was built to run at low cost, so usage to date is covered by the account's Free Tier credits ($120 remaining on 19 October); the full Bedrock agent evaluation cost $0.06. The grant funds the final stage:

Item	Estimate
GPU training of a learned crack-segmentation model (EC2 GPU instance)	$25
Graviton4 (c8g) benchmark with the OpenCV COOL build (Paid plan)	$20
Hosting the live service through judging and a 3-month pilot	$55
Bedrock usage for evaluation, demo video and judging	$10
Storage, alerts and field-test traffic	$40
Total	**$150**

6 Remaining work (to 26 October)

Deploy the calibrated quality gate and updated agent	21 Oct
Train the learned crack-segmentation model	23 Oct
Graviton4 + OpenCV COOL benchmark	24 Oct
Improve thin-crack width accuracy	24 Oct
Validation on real walls in Dhaka	25 Oct
Write-up, demo video and submission	26 Oct

Thank you for the grant and for the flexibility on the check-in.